

FUNMANAbstraction

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1 Introduction

We describe methods to stratify and abstract (de-stratify) Petrinets for compartmental models. The motivation for abstracting a Petrinet is to reduce its size, which becomes exponential in the number of stratified variables. Reducing the size of the model can have a significant impact on runtime, and it is possible to answer useful queries with the abstract model. The following sections include a background (defining the models), a description of abstraction, an approach needed to bound the abstracted models (or to perform parameter synthesis directly in a simulator), and then a comparison of simulation results for several variations of a baseline model that applies stratification, bounding, and abstraction.

2 Background

Definition 1 A Petrinet Ω is a directed graph (V, E) with vertices $V = (V_X, V_Z)$ partitioned into sets V_X of state vertices and V_Z of transition vertices.

Definition 2 Let $E(v) = E^{in}(v) \cup E^{out}(v)$ for each $v \in V$ denote edges of the form $(v, v') \in E^{out}(v)$ or $(v', v) \in E^{in}(v)$ where v is the respective source or target.

Example 1 The SIR model defines Ω by:

$$\begin{aligned} V_X &= \{v_S, v_I, v_R\} \\ V_Z &= \{v_{inf}, v_{rec}\} \\ E &= \{(v_S, v_{inf}), (v_I, v_{inf}), (v_{inf}, v_I), (v_{inf}, v_R), (v_I, v_{rec}), (v_{rec}, v_R)\} \end{aligned}$$

Definition 3 The ODE semantics Θ of the Petrinet Ω defines a tuple $(P, X, Z, \mathcal{I}, \mathcal{P}, \mathcal{X}, \mathcal{Z}, \mathcal{R})$ where

- P is a set of parameters;
- D is a set of discrete state variables, and for each $d \in D$, $\text{dom}(d)$ is a finite set of values;
- X is a set of continuous state variables;
- Z is a set of transitions;
- $\mathcal{P} : P \rightarrow \mathbb{R} \cup \mathbb{R} \times \mathbb{R}$ assigns parameters to a real number, or a pair of real numbers defining an interval;
- $\mathcal{X} : V_X \rightarrow X$ assigns a continuous variable to each state vertex;
- $\mathcal{D} : V_X \rightarrow 2^{St}$, where $St = \text{dom}(d_1) \times \dots \times \text{dom}(d_{|D|})$ is the set of all discrete states.
- $\mathcal{I} : X \rightarrow \mathbb{R}$ assigns the initial value of continuous state variables to a real number;
- $\mathcal{Z} : Z \rightarrow V_Z$ assigns transitions to transition vertices; and

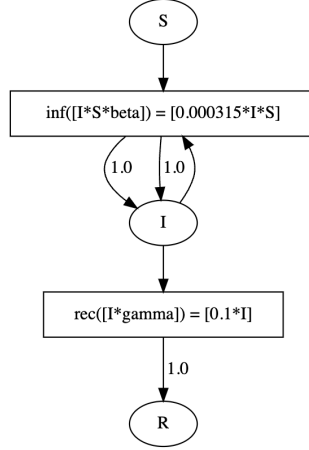


Figure 1: SIR model.

- $\mathcal{R} : \mathbf{P} \times \mathbf{V}_{\mathbf{X}} \times Z \rightarrow \mathbb{R}$ defines the rate of each transition $z \in Z$ in terms of the set of parameter vectors $\mathbf{P}$ and state variable vectors $\mathbf{X}$.

The elements of the Petrinet Ω and semantics Θ define the partial derivative

3 Stratification

Stratification is an operation that refines a model to capture finer detail about subpopulations.

Definition 4 A stratification operator $St(\Omega, \Theta, x, \{x_1, \dots, x_N\})$ defines a mapping from state variable x to a set of state variables $\{x_1, \dots, x_N\}$. The result of applying the operator is a new Petrinet Ω' and semantics Θ' , as follows:

- $X' = X \setminus \{x\} \cup \{[x, x_1], \dots, [x, x_N]\}$, where the state x is replaced by several augmented states $[x, x_i]$
- For each transition $z \in Z$, stratify z with respect to $\{x_1, \dots, x_N\}$ and each state transition $(y_1, y_2) \in st(z)$. If $(y'_1, y'_2) \in Strat((y_1, y_2), x, \{x_1, \dots, x_N\})$, then one of the following cases applies: i) $y_1, y_2 \neq x$ and $(y_1, y_2) = (y'_1, y'_2)$, the state transition is unchanged because it does not depend on x ; ii) $y_1 = x$, $y_2 \neq x$, and $(y'_1, y'_2) = ([x, x_i], y_2)$, the source state is x ; iii) $y_1 \neq x$, $y_2 = x$, and $(y'_1, y'_2) = (y_1, [x, x_j])$, the destination state is x ; or iv) $y_1, y_2 = x$, $(y'_1, y'_2) = ([x, x_i], [x, x_j])$, both states are x .
- In case iii) and iv) above, the flow from y_1 to y_2 is divided across N state transitions for each source state. The sum of the rates associated with the transitions for each source state must equal the original rate associated with (y_1, y_2) . In these cases, the new rates preserve the flow by introducing a parameter p_j for each transition and ensuring that for iii) $\sum_{j=1}^N p_j = 1$ and for iv) $\sum_{j=1}^N p_{ij} = 1$ for each $i = 1 \dots N$. These parameters represent the probability of transitioning from y_1 to $[x, x_j]$ or $[x, x_i]$ to $[x, x_j]$, respectively.
- If each state transition $(y_1, y_2) \in st(z)$ corresponds to a set of stratified state transitions $Strat((y_1, y_2), x, \{x_1, \dots, x_N\})$, then let $ST(z)$ denote the cross product of stratified state transition sets. Each element $\{(y'_{11}, y'_{21}), \dots, (y'_{1N}, y'_{2N})\} \in ST(z)$ is a set of stratified state transitions corresponding to a stratified transition z' , where $(y'_{1i}, y'_{2i}) \in Strat((y_1, y_2), x, \{x_1, \dots, x_N\})$.
- Let $E' = E \setminus E(x) \cup \bigcup_{i=1}^N E(x_i)$ denote the set edges in the stratified model, where $E(x) = \{(x, z) \in E | z \in Z\} \cup \{(z, x) \in E | z \in Z\}$, and $E(x_i) = \{(y'_{1i}, z'), (z', y'_{2i}) | (y'_{1i}, y'_{2i}) \in Strat((y_1, y_2), x, \{x_1, \dots, x_N\}), (y_1, y_2) \in st(z)\}$.

- Let $\mathcal{R}()$

Let $E(z)$ denote edges The edges E' stratified transition z'

For example, Example ?? is a stratification of the SIR model that captures two susceptible groups that have different infection rates β_1 and β_2 . Stratifications also capture how the subpopulations change and interact. For example, if the subpopulations refer to whether individuals have been vaccinated or not, then it is possible for unvaccinated individuals to become vaccinated. Similarly, individuals becoming infected may be exposed to susceptible individuals that have been vaccinated or not, each with a different infection rate. Transitions between strata can occur in conjunction with another transition (e.g., simultaneously becoming infected and vaccinated), or via “self” transitions (e.g., transitioning between susceptible groups when vaccinated). Stratification can also be partial, meaning that states like S_1 and S_2 are stratified, but I and R are not. This is useful when, for example, the recovery rate is the same for all subpopulations and there is no benefit to stratifying I or R .

We require several pieces of information to describe these cases with stratification operations. Without loss of generality, we assume that a stratification refers to a single state variable x , and that it is refined into a number of strata. In order to support different abstracts, we define stratification as a partition of a set of levels. Each stratum corresponds to one element of the partition. Under this definition, abstraction can analogously combine a subset of the levels. For example, if we stratify S with respect to attribute *vac* (for vaccinated), then it has two levels $\{T, F\}$ (for true and false). Similarly, we use attribute *age*, then it may have three levels $\{\text{young}, \text{middle}, \text{elderly}\}$. A partition of the age attribute could define $\{\{\text{young}, \text{middle}\}, \{\text{elderly}\}\}$ so that there are two strata corresponding to the non-elderly and the elderly.

Stratifying a state variable also impacts the transitions involving the variable (as either an input or output). For each transition, the stratification must define transition refinements. For example, when we refine S , we must also refine the *inf* transition, because S is an input.

In addition to a state variable, a stratification can involve a number of parameters. Each transition involving the original state variable will also be stratified, and its necessary to specify how the transition rate parameters change. We also define how a transition is stratified by the possible intra-strata transitions. For example, the age partition above defines two strata, and the stratification can define transitions between these. Each transition has an equal number of inputs and outputs, meant to preserve the flow in and out of the transition. The *inf* transition has inputs $[S, I]$ and outputs $[I, I]$. We make two assumptions to reduce the number of stratifications: 1) there is a “natural” transition as part of the transition, and 2) the total flow of a transition is distributed across the constituent stratified transitions. The *inf* transition comprises two state-to-state transitions (a one-to-one mapping from the inputs to outputs) S to I and I to I . We refer to the S to I transition as a natural transition, and the I to I , as a noop transition. Intuitively, only one individual is infected, changing state from S to I , and they are infected by an individual that is infected before and after the contact, persisting in the state I . We assume that any change of stratum co-occurring with the transition is associated with the natural transition, and that the noop state-to-state transition does not change between strata. In the stratified SIR model, this means that if we stratify both S and I , there can be state-to-state transitions S_1 to I_1 or S_1 to I_2 (where 1 and 2 refer to different strata). However, the state-to-state noop transitions will either be I_1 to I_1 or I_2 to I_2 . Because the parameters help define the transition rate, they may be different for transitions that have the state-to-state transition S_1 to I_1 versus S_1 to I_2 . Under our assumptions, there are eight possible stratified transitions of *inf* if both S and I are stratified over two strata – four possible S to I state-to-state transitions and two I to I , amounting to eight combinations. Each of these stratified transitions may use a different instance of the transition rate parameters, and stratifying *inf* in this way could require eight different values of β . Omitting β from the stratification is equivalent to stating that all instances of β are identical. In addition to the stated parameters, it is also sometimes necessary to introduce a transition probability parameter. For example, when there is a transition S_1 to I_1 and S_1 to I_2 , an individual may only transition to one of I_1 or I_2 . Without modifying the transition rate, the model will not preserve the total flow. We associate a probability distribution $P(S|I)$ to preserve the flow, and introduce parameters of the form $p_{S_1|I_1}$ and $p_{S_2|I_1}$ that correspond to the cases represented by the distribution. Following these assumptions, we are able to specify a stratification as follows.

Stratification involves defining a few concepts: stratum attributes (discrete dimensions that distinguish related model elements), stratum valuations (sets of stratum attribute value assignments), transition components (input-output pairs of states for each transition), and strata transitions (input-output pairs of strata for each transition component). The stratification operator defines how to augment the model in terms of these concepts. Intuitively, each state and transition has an associated stratum valuation that defines values of attributes consistent with them. Stratifying a transition involves stratifying each transition component and refining the associated strata transition.

Definition 5 A stratum attribute $a \in A$ is a discrete finite variable with domain $d(a) = \{v_1, v_2, \dots\}$, where each value v_i is also referred to as a level a .

Definition 6 Each stratum $s \in S$ has a valuation $V(s) \subseteq d(a_1) \times d(a_2) \times \dots$ that associates a set of levels for each stratum attribute a_i with the stratum.

We can assume that each element of a base model corresponds to a single stratum s where $V(s) = d(a_1) \times d(a_2) \times \dots$. Stratification operators partition the stratum of model elements into sub-strata.

Definition 7 A transition component $t \in T(z)$ is a pair (x_i, x_o) where $(x_i, z) \in E$ and $(z, x_o) \in E$.

The transition components express the possible one to one mappings of inputs and outputs. There may be several such mappings, but we assume for the sake of simplicity, that there is a single unique mapping. This single mapping includes at most one “natural” transition component and zero or more “noop” components. Natural transitions involve a change of state where the input and output state are not equal. The noop transitions require that the input and output states are identical. This assumption imposes a restriction on the possible transitions needed to have a single set of transition components describing a transition.

Definition 8 A stratum transition (s, s') represents a transition from one stratum valuation to another, and can be paired with a transition component to signify a simultaneous change in stratum and state.

Definition 9 Stratifying a stratum s , where $V(s) = d(a_1) \times d(a_2) \times \dots$, with respect to stratum attribute a and the partition $\mathcal{D} = [D_0|D_1|\dots]$ of the attribute’s values, will result in a set of strata, defined by the set of valuations $\{V(s) \cap_a D_i | \mathcal{D} = [D_0|D_1|\dots]\}$ where $V(s) \cap_a D_i = d(a_1) \times \dots \times d(a) \cap D_i \times \dots$.

Definition 10 Stratifying a state x associated with stratum s defines a set of state and stratum pairs $\{(x_{s_i}, s_i)\}$ where each pair corresponds to an element D_i of the partition $\mathcal{D}$.

Definition 11 Stratifying a transition component $t = (x_{in}, x_{out}) \in T(z)$ defines a set of transition component and strata transition pairs $\{((x_{in}, x_{out}), (s_i(x_{in}), s_i(x_{out})) | \mathcal{D} = [D_0|D_1|\dots], V(s_i(x_{in})) = V(s_i(x_{in})) \cap_a D_i, V(s_i(x_{out})) = V(s_i(x_{out})) \cap_a D_i\}$

Definition 12 The stratification $\text{Stratify}_Q(\Omega, \Theta) = (\Omega_Q, \Theta_Q)$ of a Petrinet and semantics with respect to a specification $Q = (x, \mathcal{D}, P', \mathcal{P}')$ defines a new Petrinet and semantics with the following properties:

- **Parameters:** For each $p \in P' \subseteq P$,
- **Parameter Values:** $\mathcal{P}'$
- **States:** For each element D_i of the partition $\mathcal{D} = [D_0|D_1|\dots]$, define a new set of states $X' = X \setminus \{x\} \cup \{x_{D_i} | \mathcal{D} = [D_0|D_1|\dots]\}$. The stratum valuation for each new state x_{D_i} is $()$
- **Transitions:** For each transition $z \in Z^{in(x)} \cup Z^{out(x)}$ (where x is an input or output), we create stratified transitions by first creating stratified state to state transitions for natural transitions and noop transitions (if present). The cross-product of these two sets of state to state transitions defines the set of stratified transitions. A natural transition will (by definition) only have x as an input or an output, but not both. Similarly, a noop transition will have x as both an input and an output. A transition

can be comprised of either one or both a natural transition and noop transition. Let $Z_{noop}(z, x)$ and $Z_{natural}(x, z)$ and $Z_{natural}(z, x)$ denote sets of stratified state to state transitions. Then,

$$\begin{aligned} Z_{noop}(z, x) &= \{((x, D_i \cap D(x)), (x, D_i \cap D(x))) | (x, z) \in Z, (z, x)Z, \mathcal{D} = [D_0|D_1|\dots]\} \\ Z_{natural}(x, z) &= \{((x, D_i \cap D(x)), (y, D(y))) | (x, z) \in Z, (z, y)Z, x \neq y, \mathcal{D} = [D_0|D_1|\dots]\} \\ Z_{natural}(z, x) &= \{((y, D(y)), (x, D_j \cap D(x))) | (y, z) \in Z, (z, x)Z, x \neq y, \mathcal{D} = [D_0|D_1|\dots]\} \end{aligned}$$

where $D(y)$ is the pre-existing stratum of y (and similarly for $D(x)$) The cross product of the non-empty sets of stratified state to state transitions will include either one or two state to state transitions that have stratum paired with each input and output.

according to the cases:

- when x is a noop transition (both an input and output), create a transition for each stratum D_i .
- when x

(where x is both an input and output of a natural transition, in cases where there is no noop transition), it is possible to transition between strata defined by the partitions D_i and D_j . For each such strata transition define $z_{D_i \rightarrow D_j}$ so that:

- $E'_{in} = E_{in} \setminus \{(z, x)\} \cup \{(z_{D_i \rightarrow D_j}, x_{D_j}) | \mathcal{D} = [D_0|D_1|\dots]\}$ if $z \in Z^{in(x)}$.
- $E'_{out} = E_{out} \setminus \{(x, z)\} \cup \{(x_{D_i}, z_{D_i \rightarrow D_j}) | \mathcal{D} = [D_0|D_1|\dots]\}$ if $z \in Z^{out(x)}$.
- $\mathcal{R}(\mathbf{p}, \mathbf{x}, z_{D_i \rightarrow D_j}) = \mathcal{R}(\mathbf{p}, \mathbf{x}, z)[\mathcal{P}'(\mathbf{p}, D_i, D_j) \setminus \mathbf{p}]$

- Initial State
- Parameter Values:
- Transition Rates:

4 Abstraction

Definition 13 An abstraction (Θ^A, Ω^A) of a Petri net and the associated semantics (Θ, Ω) that is produced by the abstraction operator A has the following properties:

- **State:** For each $x \in X$, there exists an $x' \in X^A$ where $A(x) = x'$. For each abstract state variable $x' \in X^A$, the initial value is the sum of the initial values of state variables mapped to x' by A , so that $\mathcal{I}^A(x') = \sum_{x \in X: A(x)=x'} \mathcal{I}(x)$.
- **Parameters:** For each $p \in P$, there exists a $p' \in P^A$ where $A(p) = p'$. For each abstract parameter $p' \in P^A$, the value (or interval) is the sum of all parameters mapped to p' by A , so that $\mathcal{P}^A(p') = \sum_{p \in P: A(p)=p'} \mathcal{P}(p)$.
- **Transitions:** For each $z \in Z$, there exists a $z' \in Z^A$ where $A(z) = z'$.
- **In Edges:** For each edge $(z, x) \in E_{in}$, there exists a $(z', x') \in E_{in}^A$, where $A((z, x)) = (z', x')$, $A(x) = x'$, and $A(z) = z'$.
- **Out Edges:** For each edge $(x, z) \in E_{out}$, there exists a $(x', z') \in E_{out}^A$, where $A((x, z)) = (x', z')$, $A(x) = x'$, and $A(z) = z'$.
- **Transition Rates:** For each $z' \in Z^A$,

$$\mathcal{R}^A(\mathbf{p}', \mathbf{x}', z') = \sum_{z \in Z: A(z)=z'} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) \quad (1)$$

Example 2 The abstraction (Θ^A, Ω^A) of the stratified SIR model defines (with the changed elements highlighted by “*”):

$$A = \begin{cases} S & : S_1 & * \\ S & : S_2 & * \\ I & : I \\ R & : R \\ \beta & : \beta_1 & * \\ \beta & : \beta_2 & * \\ \gamma & : \gamma \\ inf & : inf_1 & * \\ inf & : inf_2 & * \\ rec & : rec \end{cases}$$

$$\mathcal{R}^A = \begin{cases} \beta_1 S_1 I + \beta_2 S_2 I & : z_{inf} & * \\ \gamma I & : z_{rec} \end{cases}$$

In Example ??, the abstraction (Θ^A, Ω^A) maps the S_1 and S_2 state variables to the S state variable (destratifying the Petrinet). In combining the state variables, the abstract Petrinet consolidates the transitions inf_1 and inf_2 and associated rates from susceptible to infected.

Like the base model, the abstraction (Θ^A, Ω^A) defines a gradient $\nabla_{\Omega^A, \Theta^A}(\mathbf{p}^A, \mathbf{x}^A, t) = (\frac{dx_1'}{dt}, \frac{dx_2'}{dt}, \dots)^T$, in terms of Equation ??. Via Equation ??, the abstraction thus expresses the gradient by aggregating terms from the base Petrinet and semantics. It preserves the flow on consolidated transitions, but expresses the transition rates in terms of the base states. As such, the abstraction compresses the Petrinet graph structure, but at the cost of expanding the expressions for transition rates. Moreover, the transition rates refer to state variables and parameters (e.g., β_1 , β_2 , S_1 , and S_2) that are not expressed directly by the abstract Petrinet and semantics (e.g., as β and S), and by extension, the gradient. We address this in the next section.

5 Bounding Petrinets

We transform the abstraction in what we call a bounded Petrinet, so that it can refer to the abstract, and not the base, Petrinet and semantics. The same bounding can be applied to any Petrinet by controlling which parameter bounds and abstracted variables are used in constructing the bounded Petrinet. Applying bounding to abstract Petrinets allows us to summarize multiple related parameters (e.g., stratified parameters) with an interval containing the parameter values. The degenerate case of bounding a Petrinet is when the lower and upper bound on each parameter is identical, and thus the lower and upper bounds for each state variable at each time are equal.

Independent of abstraction, relaxing the parameter bounds so that the lower and upper bound are not equal allows us to perform parameter synthesis. For example, by computing the upper bound I^{ub} on the number of infected I given that $\beta \in [\beta^{lb}, \beta^{ub}]$ will allow us to conclude that any value of β within the bounds will satisfy $I^{ub} < c$ for some threshold c . FUNMAN evaluates the same type of queries by showing their negation $I \geq c$ is unsatisfiable. In constructing a modestly-larger (polynomial in the original Petrinet), bounded Petrinet, we can use off the shelf simulators to synthesize parameters.

5.1 Bounding Transformation

We present a general definition of how to bound a Petrinet and identify the special case where the Petrinet was constructed by abstraction. The key to bounding a Petrinet is to map each state variable x to a pair of state variables $[x^{lb}, x^{ub}]$. We ensure that $x^{lb}(t) \leq x(t)$ for all time points t by developing a lower bound on each contribution to the value of x , and similarly for upper bounds. The contributors to the value of x are based upon transitions $z \in Z$, where there is an in-edge $(z, x) \in E_{in}$ or out-edge $(x, z) \in E_{out}$. From Equation

??, the in-edges define positive contributions (i.e., increase the value of x) and the out-edges define negative contributions (i.e., decrease the value of x). To formulate a lower bound on x , we define the derivative $\frac{dx^{lb}}{dt}$ so that $\frac{dx^{lb}}{dt} \leq \frac{dx}{dt}$, and similarly for upper bounds (i.e., $\frac{dx}{dt} \leq \frac{dx^{ub}}{dt}$). It relies upon finding a substitution σ of the form $[x_1^*/x_1, \dots, x_n^*/x_n, p_1^*/p_1, \dots, p_m^*/p_m]$ where $*$ $\in \{lb, ub\}$ and may be different for each term.

Definition 14 A bounded Petrinet (Θ^B, Ω^B) transforms Petrinet (Θ, Ω) by replacing each element of (Θ, Ω) by a pair of elements denoting the lower and upper bound of that element (and referred to with the “lb” and “ub” superscripts). The bound transformation defines (Θ^B, Ω^B) so Theta:

- *State:* For each $x \in X$, there exists $x^{lb}, x^{ub} \in X^B$. For each $x^{lb}, x^{ub} \in X^B$, $\mathcal{I}^B(x^{lb}) = \mathcal{I}^B(x^{ub}) = \mathcal{I}(x)$.
- *Parameters:* For each $p \in P$, let $\mathcal{P}^B(p^{lb}) = \mathcal{P}^B(p^{ub}) = \mathcal{P}(p)$ if $\mathcal{P}(p)$ is a single value, or $\mathcal{P}^B(p^{lb}) = a$, $\mathcal{P}^B(p^{ub}) = b$, if $\mathcal{P}(p)$ is a pair that signifies an interval $[a, b]$.
- *Transitions:* For each transition $z \in Z$ and state variable $x \in X$, we define up to four transitions $z_{x^{lb}}^{in}$, $z_{x^{ub}}^{in}$, $z_{x^{lb}}^{out}$, and $z_{x^{ub}}^{out}$, so that Z^B is defined as follows:

$$Z^B = \{z_{x^{lb}}^{in}, z_{x^{ub}}^{in} | (z, x) \in E_{in}\} \cup \{z_{x^{lb}}^{out}, z_{x^{ub}}^{out} | (x, z) \in E_{out}\}$$

- *In Edges:* For each edge $(z, x) \in E_{in}$, there is a pair of transitions $(z_{x^{lb}}^{in}, x^{lb}), (z_{x^{ub}}^{in}, x^{ub}) \in E_{in}^B$.
- *Out Edges:* For each edge $(x, z) \in E_{out}$, there is a pair of transitions $(x^{lb}, z_{x^{lb}}^{out}), (x^{ub}, z_{x^{ub}}^{out}) \in E_{in}^B$.
- *Transition Rates:* For each transition in $z \in Z^B$, the rate $\mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, z)$ defines the rate of z , expressed in terms of a vector of parameters $\mathbf{p}^B$ (corresponding to P^B) and a vector of state variables $\mathbf{x}^B$ (corresponding to X^B). The rate for each transition in Z^B depends on finding a substitution σ (mapping $\mathbf{p}$ to $\mathbf{p}^B$ and $\mathbf{x}$ to $\mathbf{x}^B$) that obeys the following cases:

$$\begin{aligned} \mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, z_{x^{lb}}^{in}) &= \arg \min_{\sigma} (\sigma \circ ([x^{lb}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z))) \\ \mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, z_{x^{lb}}^{out}) &= \arg \max_{\sigma} (\sigma \circ ([x^{lb}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z))) \\ \mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, z_{x^{ub}}^{in}) &= \arg \max_{\sigma} (\sigma \circ ([x^{ub}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z))) \\ \mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, z_{x^{ub}}^{out}) &= \arg \min_{\sigma} (\sigma \circ ([x^{ub}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z))) \end{aligned}$$

The $\circ$ operator applies a substitution to an expression and returns the resulting expression. In the rates, we first substitute the lower bound x^{lb} for x , and find a substitution σ for the remaining state variables and parameters, and similarly for upper bounds.

The rates capture how each transition either effects x positively by flowing in, or negatively by flowing out. Propagating a lower bound involves minimizing the positive flow in, and maximizing the negative flow out. Propagating upper bounds follows similar rationale, maximizing flow in, and minimizing flow out.

Equations ?? to ?? show how we derive the lower bound $\frac{dx^{lb}}{dt}$. The primary intuition is that there exists a substitution σ that minimizes (or maximizes) each expression in brackets, and that by moving the choice of substitution inward we allow each sub-expression a different choice of σ . This is a type of independence assumption that allows us to symbolically minimize (or maximize) each rate by choosing an appropriate substitution.

$$\frac{dx}{dt} = \sum_{z \in Z^{in}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) - \sum_{z \in Z^{out}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) \quad (2)$$

$$\geq \arg \min_{\sigma} \sigma \circ \left[\sum_{z \in Z^{in}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) - \sum_{z \in Z^{out}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) \right] \quad (3)$$

$$\geq \arg \min_{\sigma} \sigma \circ \left[\sum_{z \in Z^{in}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) \right] - \arg \max_{\sigma} \sigma \circ \left[\sum_{z \in Z^{out}(x)} \mathcal{R}(\mathbf{p}, \mathbf{x}, z) \right] \quad (4)$$

$$\geq \sum_{z \in Z^{in}(x)} \arg \min_{\sigma} \sigma \circ [\mathcal{R}(\mathbf{p}, \mathbf{x}, z)] - \sum_{z \in Z^{out}(x)} \arg \max_{\sigma} \sigma \circ [\mathcal{R}(\mathbf{p}, \mathbf{x}, z)] \quad (5)$$

$$\geq \sum_{z \in Z^{in}(x)} \arg \min_{\sigma} \sigma \circ [[x^{lb}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z)] - \sum_{z \in Z^{out}(x)} \arg \max_{\sigma} \sigma \circ [[x^{lb}/x] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, z)] \quad (6)$$

$$= \frac{dx^{lb}}{dt} \quad (7)$$

For example, in the stratified SIR model from Example ??, we can derive a lower bound for I , as listed in Equations ?? to ??, and the choice of substitutions listed in Equations ?? to ??. As in σ_{rec} , the substitutions must include I^{lb}/I whenever I is a rate term because we are defining $\frac{dI^{lb}}{dt}$.

$$\frac{dI}{dt} = \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_1) + \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_2) - \mathcal{R}(\mathbf{p}, \mathbf{x}, rec) \quad (8)$$

$$\geq \sigma_1 \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_{1I^{lb}})) + \quad (9)$$

$$\sigma_2 \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_{2I^{lb}})) - \quad (10)$$

$$\sigma_3 \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, rec_{I^{lb}})) \quad (11)$$

$$= \frac{dx^{lb}}{dt} \quad (12)$$

$$\sigma_1 = [S_1^{lb}/S_1, \beta_1^{lb}/\beta_1] \quad (13)$$

$$\sigma_2 = [S_2^{lb}/S_2, \beta_2^{lb}/\beta_2] \quad (14)$$

$$\sigma_3 = [\gamma^{ub}/\gamma] \quad (15)$$

Applying the substitutions to the rate terms, $\frac{dI}{dt}$ and $\frac{dI^{lb}}{dt}$ are simplified to the expressions in Equations ?? to ??.

$$\frac{dI}{dt} = \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_1) + \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_2) - \mathcal{R}(\mathbf{p}, \mathbf{x}, rec) \quad (16)$$

$$= S_1 I \beta_1 + S_2 I \beta_2 - I \gamma \quad (17)$$

$$\frac{dI^{lb}}{dt} = [S_1^{lb}/S_1, \beta_1^{lb}/\beta_1] \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_{1I^{lb}})) + \quad (18)$$

$$[S_2^{lb}/S_2, \beta_2^{lb}/\beta_2] \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, inf_{2I^{lb}})) - \quad (19)$$

$$[\gamma^{ub}/\gamma] \circ ([I^{lb}/I] \circ \mathcal{R}(\mathbf{p}, \mathbf{x}, rec_{I^{lb}})) \quad (20)$$

$$= S_1^{lb} I^{lb} \beta_1^{lb} + S_2^{lb} I^{lb} \beta_2^{lb} - I^{lb} \gamma^{ub} \quad (21)$$

5.2 Bounding Abstracted Petrinets

We previously described how abstracted Petrinets contain state variables and parameters from the previously stratified Petrinet in the transition rates. For example, if $A(S_1) = S$ and $A(S_2) = S$ (S_1 and S_2 are stratified variables represented by S in the abstraction), the transition rate associated with the inf transition is

$$\mathcal{R}^A(\mathbf{p}^A, \mathbf{x}^A, inf) = \beta_1 S_1 I + \beta_2 S_2 I$$

By construction, we know that $S_1 + S_2 = S$. Applying this equivalence to the rate law as a substitution results in:

$$\begin{aligned} [S - S_1/S_2] \circ \mathcal{R}^A(\mathbf{p}^A, \mathbf{x}^A, inf) &= \beta_1 S_1 I + \beta_2 (S - S_1) I \\ &= \beta_1 S_1 I + \beta_2 S I - \beta_2 S_1 I \end{aligned}$$

Furthermore, when bounding this rate to determine $\mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, inf_{I_{lb}}^{in})$, it is possible to substitute the same term $\min(\beta_1, \beta_2)$ for both parameters β_1 and β_2 :

$$\begin{aligned} \mathcal{R}^B(\mathbf{p}^B, \mathbf{x}^B, inf_{I_{lb}}^{out}) &= [\min(\beta_1, \beta_2)/\beta_1, \min(\beta_1, \beta_2)/\beta_2, S^{lb}/S, I/I^{lb}] \circ ([S - S_1/S_2] \circ \mathcal{R}^A(\mathbf{p}^A, \mathbf{x}^A, inf)) \\ &= \min(\beta_1, \beta_2) S_1 I + \min(\beta_1, \beta_2) S^{lb} I^{lb} - \min(\beta_1, \beta_2) S_1 I \\ &= \min(\beta_1, \beta_2) S^{lb} I^{lb} \\ &= \beta^{lb} S^{lb} I^{lb} \end{aligned}$$

where we use a new parameter β^{lb} and define $\mathcal{P}^B(\beta^{lb}) = \min(\beta_1, \beta_2)$. A similar argument can be made for the upper bound using $\max(\beta_1, \beta_2)$.

By introducing the bounded parameters, we no longer rely upon the base state variables or parameters. However, in tracking the effect of the bounded parameters, the bounded abstraction must also track bounded rates and bounded state variables. The resulting bounded abstraction thus over-approximates the abstraction and base model, wherein we can derive bounds on the state variables at each time, which may correspond to a larger (hence over-approximation) set of state trajectories.

5.3 Bounded Petrinet Example

Example 3 The bounded abstraction (Θ^B, Ω^B) of the stratified SIR model defines:

$$\begin{aligned}
E^B &= ((S^{lb}, inf_{S_{lb}}^{out}), (S^{ub}, inf_{S_{ub}}^{out}), (I^{lb}, inf_{I_{lb}}^{out}), (I^{ub}, inf_{I_{ub}}^{out}), (inf_{I_{lb}}^{in} I^{lb}), (inf_{I_{ub}}^{in} I^{ub}), \\
&\quad (I^{lb}, rec_{I_{lb}}^{out}), (I^{ub}, rec_{I_{ub}}^{out}), (rec_{R_{lb}}^{in}, R^{lb}), (rec_{R_{ub}}^{in}, R^{ub})) \\
P^B &= \{\beta^{lb}, \beta^{ub}, \gamma^{lb}, \gamma^{ub}\} \\
X^B &= \{S^{lb}, S^{ub}, I^{lb}, I^{ub}, R^{lb}, R^{ub}\} \\
Z^B &= \{inf_{S_{lb}}^{out}, inf_{S_{ub}}^{out}, inf_{I_{lb}}^{out}, inf_{I_{ub}}^{out}, inf_{I_{lb}}^{in}, inf_{I_{ub}}^{in}, rec_{I_{lb}}^{out}, rec_{I_{ub}}^{out}, rec_{R_{lb}}^{in}, rec_{R_{ub}}^{in}\} \\
\mathcal{I}^B &= \begin{cases} 0.9 & : S^{lb} \\ 0.9 & : S^{ub} \\ 0.1 & : I^{lb} \\ 0.1 & : I^{ub} \\ 0.0 & : R^{lb} \\ 0.0 & : R^{ub} \end{cases} \\
\mathcal{P}^B &= \begin{cases} 1e-7 & : \beta^{lb} \\ 2e-7 & : \beta^{ub} \\ 1e-5 & : \gamma^{lb} \\ 1e-5 & : \gamma^{ub} \end{cases} \\
\mathcal{R}^B &= \begin{cases} \beta^{ub} S^{lb} I^{ub} & : inf_{S_{lb}}^{out} \\ \beta^{lb} S^{ub} I^{lb} & : inf_{S_{ub}}^{out} \\ \beta^{ub} S^{ub} I^{lb} & : inf_{I_{lb}}^{out} \\ \beta^{lb} S^{lb} I^{ub} & : inf_{I_{ub}}^{out} \\ \beta^{lb} S^{lb} I^{lb} & : inf_{I_{lb}}^{in} \\ \beta^{ub} S^{ub} I^{ub} & : inf_{I_{ub}}^{in} \\ \gamma^{ub} I^{lb} & : rec_{I_{lb}}^{out} \\ \gamma^{lb} I^{ub} & : rec_{I_{ub}}^{out} \\ \gamma^{lb} I^{lb} & : rec_{R_{lb}}^{in} \\ \gamma^{ub} I^{ub} & : rec_{R_{ub}}^{in} \end{cases}
\end{aligned}$$

6 SIR Example Results

Results are available in this notebook.

7 SIRHD Example Results

We measured the time to simulate various formulations of the SIRHD model to highlight the effects of stratification, abstraction, and bounding. We defined a series of stratification, abstraction, and bounding operations as follows. Starting with the base model (Ω, Θ) , we either bound the model $(\Omega^B, \Theta^B) = \text{Bound}(\Omega, \Theta)$, or stratify the model $(\Omega^S, \Theta^S) = \text{Stratify}(\Omega, \Theta)$. Each stratified model (Ω^S, Θ^S) is stratified again differently $(\Omega^{S'}, \Theta^{S'}) = \text{Stratify}(\Omega^S, \Theta^S)$, bounded $(\Omega^B, \Theta^B) = \text{Bound}(\Omega^S, \Theta^S)$, or abstracted $(\Omega^A, \Theta^A) = \text{Abstract}(\Omega^A, \Theta^A)$. Each abstracted model (Ω^A, Θ^A) is bounded $(\Omega^B, \Theta^B) = \text{Bound}((\Omega^A, \Theta^A))$, or abstracted again $(\Omega^{A'}, \Theta^{A'}) = \text{Abstract}(\Omega^A, \Theta^A)$. While it is also possible to stratify an abstracted model, we don't explore this operation here.

Figure ?? describes how we developed models from the SIRHD base model. We used several stratifications, and then applied abstractions that reversed the stratifications. Each abstraction was bounded so that we could simulate the model. We organized the models in the fashion so that we could demonstrate the time to simulate

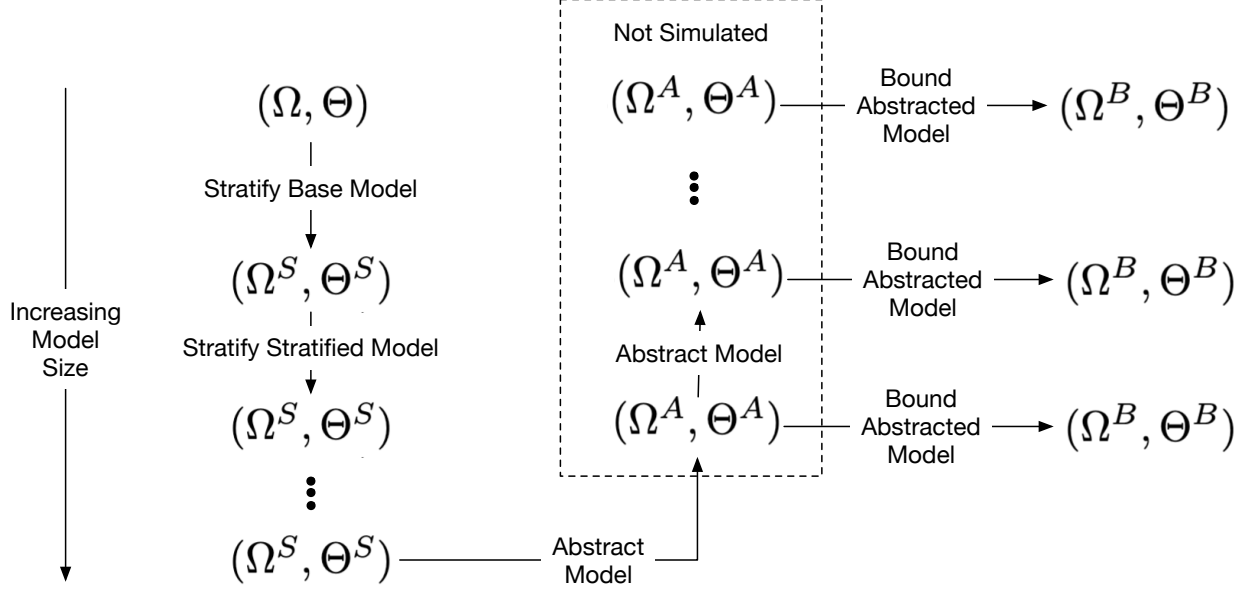


Figure 2: Conceptual relationships between model formulations. The base model can be stratified a number of times. The final stratified model is abstracted to reverse the stratifications. Each abstracted model is bounded so that it can be simulated. The dashed box indicates which model are not simulated.

each model. In the figure, the models are aligned vertically to indicate which models represent the same level of detail. For example, the base model and the most-abstract abstraction have the same number of states, transitions, and parameters. Likewise, reversing the last stratification with the first abstraction results in a model that is the same size as the one prior to the last stratification. Bounding an abstracted model will increase its size polynomially, whereas stratifying it will increase its size exponentially. The runtime results tables arrange the models in a similar fashion. The first column of results increases the model size by stratification from the top to the bottom, and the second column decreases the model size from the top to the bottom.

Table ?? lists runtime results in seconds to simulate each SIRHD model variation. Each instance seeks to check a constraint that assesses whether over 200 days the number of infected is no more than 30% of the population ($N = 1.5e9$). We use 15 stratifications on the model, three for each of the state variables S , I , R , H and D . The three stratifications involve stratifying each state variable into vaccinated and unvaccinated groups (e.g., S_{vac} and S_{unvac}), the vaccinated group into age groups (e.g., $S_{vac,0}$, $S_{vac,1}$, $S_{vac,2}$, ...), and the unvaccinated group into age groups (e.g., $S_{unvac,0}$, $S_{unvac,1}$, $S_{unvac,2}$, ...). We report results for either three or five age groups. The model index 0-15 refers to successive stratifications of the base model 0 in the Stratified column. Similarly, the Bounded Abstracted column refers to successive abstractions of the models, where model 14 is the abstraction of model 15 in the Stratified column, and model 13 is the abstraction of model 14 in the Bounded Abstracted column.

We construct the most abstract model 0 from a series of stratifications and abstractions. It is an over-approximation of the most stratified model that may allow us to prove the constraint is satisfied. The abstract bounded model represents lower and upper bounds on the number of infected, and if the upper bound is less than 30% of N (checked through simulation) then we do not need to simulate the most stratified model 15 (our reference model formulation). The bounded abstract model 0 can be inconclusive if 30% of N falls between the lower and upper bounds on the number of infected. By considering successively less abstract models (i.e., proceeding from model 0 to model 1), we expect the lower and upper bounds to tighten. The bolded time in the Bounded Abstracted model column is the runtime of the first model formulation (starting from model 0) where we can prove that the upper bound on the number of infected is less than 30% of N . Therefore, if we

Model	3 Age Levels		5 Age Levels	
	Stratified	Bounded Abstracted	Stratified	Bounded Abstracted
0	0.57	5.26	0.54	5.35
1	0.76	7.18	0.69	6.93
2	1.33	5.92	2.86	7.43
3	2.18	9.93	6.29	22.98
4	3.34	9.50	6.17	27.63
5	3.88	14.25	11.46	52.81
6	5.36	22.84	19.47	108.55
7	7.31	23.12	22.97	114.84
8	7.13	28.87	26.41	143.46
9	8.14	34.59	39.39	200.15
10	9.25	32.94	36.55	209.17
11	12.36	41.55	49.47	245.51
12	12.96	50.17	58.79	319.67
13	11.49	56.79	61.23	292.44
14	13.73	67.72	66.92	361.31
15	15.17	-	83.80	-

Table 1: Runtime in seconds to simulate each model formulation. Model 0 in the Stratified column is the base model. Models 1-15 in the Stratified column are successive stratifications. From model 14 to 0 in the Bounded Abstracted column each is a successive abstraction. The bolded number in the Stratified column is the fully stratified model that we assume is the starting point for analysis. The bolded number in the Bounded Abstracted column is the smallest model that allows us to confirm the model constraint is satisfied.

need to simulate Bounded Abstracted models 0 through 3, we require 28.29 and 42.69 seconds, respectively, for either three or five age groups. Compared to the time to simulate the largest Stratified model 15, it takes 15.17 or 82.80 seconds respectively. The savings due to abstraction reduces runtime by one half when there are five age groups, but doubles the runtime when there are three age groups. This illustrates the trade-off between simulating several abstract models versus one large stratified model. When the abstract models can capture the important model dynamics using bounds and the number of collapsed stratification levels is large, then abstraction is a win.